FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 · SUB-STRAND 3.1: ANIMAL NUTRITION	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION ASSESSMENT — Animal Nutrition (Sub-strand 3.1)

DRIVING QUESTION: How do the structures animals use to get food — from an aphid's needle-like mouthparts to a flamingo's filter beak to your own teeth and digestive system — explain how each animal gets the nutrients it needs to survive?

WHAT THIS TASK IS ASKING YOU TO DO:
You have spent 10 lessons investigating how animals obtain and process food, starting from a puzzling observation on a maize farm in Kakamega. You watched grasshoppers chew ragged holes in maize leaves, aphids silently tap into plant stems, and a mosquito pierce skin to draw blood — all on the same afternoon. At Lake Nakuru, you compared a flamingo's sideways-sweeping beak with a fish eagle's hooked, tearing beak. Now it is time to bring all of that thinking together into one well-organised, evidence-based explanation.

Your Final Explanation should answer the driving question fully by connecting: (1) the external structures animals use to capture and process food, (2) the internal digestive structures that break food down, and (3) the relationship between an animal's diet, its structures, and its ecological role. Use specific examples from the Kakamega farm and Lake Nakuru phenomenon, as well as examples from your lessons, to support every claim you make.

HOW TO USE THIS DOCUMENT:
Read each section prompt carefully. Study the exemplar response provided — it shows the quality and depth of thinking expected. Then write your own response in the space provided. Your teacher will score your work using the rubric at the end of this document.

TIPS FOR STRONG ANSWERS:
- Make a clear CLAIM, support it with EVIDENCE, and explain your REASONING (CER framework).
- Use precise scientific vocabulary: mandibles, proboscis, radula, lamellae, peristalsis, enzyme, villi, absorption, adaptation.
- Connect every answer back to the anchoring phenomenon — the Kakamega farm insects and/or the Lake Nakuru birds.
- Write in full sentences and organise your ideas logically.

Section 1 — Mouthpart Adaptations: Structure Matches Diet	
Explain how the mouthparts of the grasshopper, aphid, and mosquito you encountered in the Kakamega phenomenon are each adapted to their specific diets. In your answer: (a) name and describe the key structural feature of each insect's mouthparts, (b) explain how that structure makes it possible for the insect to obtain its food, and (c) explain why each insect CANNOT feed the way the other two do. Then extend your explanation to include either the flamingo or the fish eagle at Lake Nakuru.	The three insects at the Kakamega maize farm each use completely different mouthpart structures because they eat completely different foods, and this is a perfect example of how structure determines function. The grasshopper has strong, hardened mandibles — paired, opposing jaws that move side-to-side. These mandibles are adapted to bite off and grind solid plant material like maize leaves. The chewing surfaces are ridged, which helps break tough cellulose cell walls so the grasshopper can access nutrients inside the plant cells. A grasshopper cannot suck blood or extract sap because its mandibles cannot pierce skin or plant phloem tubes — they are built to crush, not to penetrate. The aphid has a needle-like stylet (a modified proboscis) that it inserts precisely between plant cells and into the phloem sieve tubes of the maize stem. Phloem sap is already under pressure, so once the stylet reaches the phloem, the sugary liquid flows into the aphid without much effort. This explains why the leaves wilt — the aphid drains the plant's own transport system. An aphid cannot chew leaves because its mouthparts have no cutting or grinding surfaces; and it cannot pierce vertebrate skin because its stylet is calibrated for the thin walls of plant tissue, not tough skin with pain receptors. The mosquito also has a proboscis, but it is more complex — it contains six needle-like stylets enclosed in a flexible sheath. Two stylets saw through the skin, two hold the tissue apart, one injects anticoagulant saliva (which is why the bite does not hurt at first), and one draws up blood. This precision drilling-and-pumping system allows the mosquito to reach blood capillaries just beneath the skin. A mosquito cannot chew maize leaves or tap plant sap efficiently because its mouthparts lack the strength to grind solid tissue and its proboscis is designed for the specific resistance of vertebrate skin, not plant stem walls. At Lake Nakuru, the flamingo extends this idea further. A flamingo's beak is uniquely bent downward and contains rows of comb-like lamellae along the edges. The flamingo sweeps its beak upside-down through shallow, alkaline water, and the lamellae act like a sieve, filtering out tiny cyanobacteria (blue-green algae, particularly Arthrospira, the spirulina that makes Lake Nakuru famous) while water is pumped out by the tongue. This filter-feeding adaptation means the flamingo can harvest millions of microscopic organisms in a single sweep — food that would be impossible to collect by chewing or piercing. In every case, the shape of the mouthpart or beak is a direct solution to the challenge of obtaining a specific food from a specific environment.

Section 2 — Digestion: Breaking Down Food from the Inside	
Once food enters an animal's body, it must be broken down before nutrients can be used. Using the human digestive system as your main example: (a) distinguish between mechanical and chemical digestion and give one example of each, (b) trace the journey of a bite of ugali (made from maize flour) from the mouth to the small intestine, naming the organs, secretions, and	Digestion is the process of breaking large, complex food molecules into small, simple molecules that can be absorbed into the blood and used by body cells. There are two types: mechanical digestion and chemical digestion, and they work together. Mechanical digestion is the physical breaking of food into smaller pieces without changing its chemical structure. In the mouth, your teeth perform mechanical digestion — incisors bite off a piece of ugali, canines grip it, and premolars and molars grind it into a soft bolus (ball of chewed food). In the stomach, powerful muscular churning further breaks food into a liquid called chyme. Mechanical digestion increases the surface area of food, which makes chemical digestion faster and more efficient. Chemical digestion uses enzymes — biological catalysts — to break the chemical bonds in large molecules. When you chew ugali, your salivary glands release saliva containing the enzyme salivary amylase (also called ptyalin). Amylase begins breaking down the starch in the maize flour into shorter chains of sugars called maltose. This is why ugali starts to taste slightly sweet if you chew it for a long time. The bolus is then swallowed and pushed down the oesophagus by waves of muscular contractions called

enzymes involved at each stage, and (c) explain why digestion is necessary — what would happen if large food molecules tried to enter the bloodstream directly?	peristalsis — this is not digestion itself, but the transport system that moves food along. In the stomach, the ugali bolus meets gastric juice, which contains hydrochloric acid (HCl) and the enzyme pepsin. HCl creates a strongly acidic environment (pH 1–2) that kills most bacteria and activates pepsin. However, since ugali is mostly starch and not protein, pepsin has little to act on here. The stomach churns the food into chyme over 2–4 hours, then releases it in small amounts into the small intestine. The small intestine is where most chemical digestion is completed and where absorption happens. In the first section (duodenum), bile from the liver (stored in the gallbladder) is released. Bile is not an enzyme — it is an emulsifier that breaks fat globules into tiny droplets, increasing the surface area for fat-digesting enzymes. Pancreatic juice from the pancreas arrives simultaneously, containing pancreatic amylase (to finish starch digestion), lipase (to digest fats into fatty acids and glycerol), and protease enzymes like trypsin (to digest any proteins). By the time the ugali reaches the ileum (the longest part of the small intestine), starch has been fully converted to glucose. Digestion is absolutely necessary because glucose, fatty acids, and amino acids are small enough to pass through cell membranes into the blood — but starch, fat, and protein molecules are far too large. Think of it like trying to push a whole maize cob through a window screen: you must first grind it into tiny particles that can pass through the holes. If undigested starch entered the blood directly, the immune system would treat it as a foreign invader, and cells could not use it for energy anyway, because cells are only equipped to process simple sugars like glucose.

Section 3 — Absorption and the Role of the Small Intestine	
Explain how the small intestine is structurally adapted to absorb nutrients efficiently. In your answer: (a) describe at least THREE structural features of the small intestine and explain how each feature increases absorption, (b) explain how glucose and fatty acids take different routes into the body after absorption, and (c) connect absorption back to the phenomenon — once a mosquito's blood meal or an aphid's phloem sap is digested, what nutrients are released and why does the animal need them?	The small intestine is remarkably well-adapted for absorption — it is the main site where digested nutrients move from the gut into the body. Three key structural features explain why it is so efficient. First, the small intestine is very long — in an adult human it is about 6–7 metres. This length means food spends a long time inside and has a large contact area with the intestinal wall, giving more opportunity for nutrients to be absorbed before the food moves on. Second, the inner wall of the small intestine is folded into millions of tiny finger-like projections called villi (singular: villus). Each villus increases the surface area available for absorption. Inside each villus is a network of blood capillaries and a central lymph vessel called a lacteal. Nutrients that are absorbed pass through the thin walls of the villi directly into these vessels. Third, each villus is itself covered in even tinier projections called microvilli, which together form the brush border. The microvilli increase the surface area even further — scientists estimate that the total absorptive surface of a human small intestine, when all folds, villi, and microvilli are counted, is approximately 250 square metres. That is roughly the size of a tennis court, all folded inside your abdomen. Glucose and amino acids are small, water-soluble molecules. They are absorbed directly into the blood capillaries inside the villi by a combination of diffusion and active transport (which requires energy to move molecules against a concentration gradient). From the capillaries, they travel in the blood through the hepatic portal vein to the liver, where glucose is either used immediately, stored as glycogen, or converted to fat.

	Fatty acids and glycerol (from fat digestion) are not water-soluble, so they cannot enter blood capillaries directly. Instead, they are absorbed into the lacteals (lymph vessels) inside the villi and travel through the lymphatic system before eventually entering the bloodstream near the heart. Connecting this to the phenomenon: when a mosquito digests a blood meal, it breaks down haemoglobin and plasma proteins into amino acids. Female mosquitoes use these amino acids to synthesise proteins for egg development — this is why only female mosquitoes bite. When an aphid digests phloem sap, it absorbs glucose and amino acids for energy (cellular respiration) and growth. In both cases, the fundamental nutrients released — glucose, amino acids, fatty acids — are the same building blocks that all animals need, regardless of how dramatically different their feeding structures are. The mouthparts and beaks change; the biochemical needs do not.

Section 4 — Adaptation, Diet, and Ecological Role	
Bring together everything you have learned to answer this synthesis question: How does an animal's feeding structure reflect its ecological role and the environment it lives in? In your answer: (a) compare ANY TWO animals from the phenomenon (or from your lessons) that live in the same environment but eat different foods, explaining how their different mouthparts or beaks prevent direct competition, (b) explain what would happen to an animal population if its food source disappeared and its mouthparts could not adapt, and (c) reflect on what the feeding structures of ALL the animals in this unit — from insects to birds to humans — have in common at the most fundamental level.	An animal's feeding structures are not random — they are the result of long-term adaptation to a specific food source in a specific environment, and they directly determine what ecological role an animal plays in its ecosystem. Consider the two birds at Lake Nakuru: the flamingo and the fish eagle. They share the same lake environment, but they eat completely different foods and have completely different feeding structures — and this is precisely why they can coexist without competing directly. The flamingo's downward-bent beak with lamellae is built exclusively to filter microscopic cyanobacteria from alkaline water. It could not catch a fish even if it tried, because its beak has no gripping or tearing capability. The fish eagle, by contrast, has a sharp, strongly hooked beak for tearing flesh and powerful talons for gripping slippery fish. It could not filter microorganisms from water — its beak has no filtering apparatus. Because they eat at completely different levels of the food web (flamingos are primary consumers of algae; fish eagles are tertiary consumers of fish), they occupy different ecological niches. This principle — that different feeding structures reduce competition between species sharing a habitat — is called resource partitioning. If a food source disappeared, an animal with highly specialised mouthparts would be in serious danger. Lake Nakuru has experienced pollution events and water level changes that have reduced cyanobacteria populations. When this happens, flamingo numbers drop dramatically — the birds must fly hundreds of kilometres to Lake Bogoria or Lake Elementaita to find food. Because a flamingo's beak cannot switch to catching fish or chewing grass, the entire population depends on that one food source. Overspecialisation makes a species efficient in good times but vulnerable in bad times. A generalist feeder like the crow or the human — with mouthparts (or teeth) that can handle a wide variety of foods — can survive environmental change more easily. At the most fundamental level, all the feeding structures we have studied — the grasshopper's mandibles, the aphid's stylet, the mosquito's proboscis, the flamingo's lamellae, the fish eagle's hooked beak, and your own set of incisors, canines, premolars, and molars — are solving the same problem: how to get nutrients from the environment into the body in a form that cells can use. Every structure, no matter how different in appearance, is an answer to the same biological question. The diversity of feeding structures across the animal kingdom is simply the diversity of solutions that evolution has produced to meet that one universal need: energy and materials for survival, growth, and reproduction.

Section 5 — Revisiting the Phenomenon: Your Complete Answer	
Return to the original driving	When the Grade 10 learner stood on that maize farm in Kakamega watching grasshoppers, aphids, and mosquitoes all feeding at

question and the anchoring phenomenon one final time. Write a complete, well-organised response (at least three paragraphs) that answers: 'How do the structures animals use to get food — from an aphid's needle-like mouthparts to a flamingo's filter beak to your own teeth and digestive system — explain how each animal gets the nutrients it needs to survive?' Your answer must: (a) directly reference at least THREE animals from the phenomenon, (b) connect external feeding structures to internal digestion and absorption, and (c) state a unifying principle that applies to ALL animals, including humans.	the same time, they were witnessing one of biology's most important principles in action: structure determines function, and function determines survival. Each insect's feeding structure is a precision tool shaped by its diet. The grasshopper's mandibles grind solid plant material, tearing through maize leaves to access the nutrients locked inside plant cells. The aphid's needle-like stylet bypasses the tough leaf altogether, tapping directly into the plant's phloem to steal the sugary sap being transported through the stem — which is why the leaves wilt without being physically damaged. The mosquito's six-stylet proboscis is a marvel of biological engineering, allowing it to saw through skin, prevent clotting, and pump blood upward — giving it access to a completely different nutrient source inside another animal's body. Despite how different these tools look, all three insects ultimately absorb the same basic molecules — glucose, amino acids, and other nutrients — into their bodies for energy and growth. At Lake Nakuru, the same principle scales up dramatically. The flamingo's curved, lamellae-lined beak is not a beak in the conventional sense — it is a biological filter, designed to harvest billions of cyanobacteria from alkaline lake water. The fish eagle's hooked beak and powerful feet are designed for a completely different task: catching, gripping, and tearing apart fish. Both birds live on the same lake, but their radically different structures mean they feed at different levels of the food web and never compete for the same food. Inside both birds, however, the same digestive process occurs: mechanical and chemical breakdown of food, enzyme action to split large molecules, and absorption through the gut wall into the blood. The internal machinery of digestion is far more similar across the animal kingdom than the external feeding tools suggest. The unifying principle that ties all of this together — from an aphid on a maize stem in Kakamega to a flamingo at Lake Nakuru to a student eating ugali and sukuma wiki for lunch — is this: every animal's feeding structure is an adaptation that solves the problem of getting specific nutrients from its specific environment into its body. External structures (mouthparts, beaks, teeth) determine what food can be obtained and processed. Internal structures (stomach, intestines, enzymes, villi) determine how that food is broken down and absorbed. Together, these systems ensure that no matter what form the food takes on the outside — solid leaf, pressurised plant sap, blood, microscopic algae, or a plate of githeri — the end result inside the animal is always the same: simple molecules like glucose, fatty acids, and amino acids that cells can use to stay alive. That is the answer to why a mosquito can drink your blood but a grasshopper cannot: not because one is stronger or smarter, but because each has a body built — from beak to gut — for one specific solution to the universal challenge of obtaining nutrition.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Scientific Accuracy — Structures & Adaptations	All mouthpart/beak structures are named correctly (mandibles, stylet, proboscis, lamellae, etc.) and their functions are explained with precise detail. Structural features are accurately linked to specific diets with no factual errors.	Most structures are named and linked to function correctly. One or two minor inaccuracies or missing details, but the overall explanation is scientifically sound.	Some structures are named but functions are vague or partially incorrect. Key vocabulary (e.g., proboscis, lamellae, villi) is missing or misused. Explanations rely on general statements rather than specific biological mechanisms.
Understanding of Digestion and Absorption	Clearly distinguishes mechanical and chemical digestion with accurate examples. Correctly traces nutrients through all major organs, names relevant enzymes (amylase, pepsin, lipase), and accurately explains the different absorption routes for glucose vs. fatty	Digestion and absorption are explained with reasonable accuracy. Most organs and enzymes are correctly identified. The distinction between glucose and fatty acid absorption routes may be simplified but is not wrong. Villi are mentioned with a basic explanation.	Digestion is described in general terms only. Organs may be listed without explaining their function. Enzymes are rarely named or are confused. Absorption is mentioned but the role of villi or different transport routes is not explained.

	acids. Villi/microvilli are described with their role explained.		
Use of Evidence from the Phenomenon	At least three animals from the Kakamega/Lake Nakuru phenomenon are used as specific evidence in multiple sections. Explanations explicitly connect observations (e.g., wilting leaves, painless bite, flamingo sweeping) to biological mechanisms. The phenomenon is revisited meaningfully in the final section.	At least two animals from the phenomenon are used as evidence. Connections between observations and mechanisms are made but may be brief or stated without full explanation. The final section references the phenomenon.	References to the phenomenon are vague or limited to one animal. Evidence from the Kakamega farm or Lake Nakuru is mentioned but not used to support specific claims. The final section does not clearly revisit the phenomenon.
CER Structure — Claim, Evidence, Reasoning	Every major point follows the CER structure: a clear claim is made, specific evidence (named animals, named structures, named processes) is provided, and reasoning explains WHY the evidence supports the claim. Arguments are logically organised across all five sections.	Most points include a claim and evidence. Reasoning is present but sometimes implied rather than fully stated. The overall argument is logical, though one or two sections may lack a clear link between evidence and conclusion.	Claims are made without sufficient evidence, or evidence is listed without reasoning. Sections read as a list of facts rather than a connected argument. It is often unclear why the stated evidence supports the claim.
Unifying Principle and Synthesis	The final section (and ideally earlier sections) articulates a clear, accurate unifying principle that applies to all animals (e.g., structure determines function; all feeding adaptations solve the same fundamental problem of obtaining nutrients). The principle is used to connect external mouthpart diversity to shared internal digestion processes.	A unifying principle is stated in the final section, connecting feeding structures to digestion. The principle is reasonable but may be stated simply or not fully applied across all animals discussed.	No unifying principle is stated, or the stated principle is too vague to be meaningful (e.g., 'all animals need food'). The response does not connect external feeding structures to internal digestion or to a broader biological idea.